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Calibrated Weak Supervision for Post-Harvest Burned-Cropland Mapping Under Label Scarcity

Raunak Bhagate **Maitri Polisetty** **R I Minu**
rb7314@srmist.edu.in

SRM Institute of Science & Technology

Abstract

Crop-residue burning after harvest is a major source of air pollution in Punjab, India, but many of the burned fields are too small for the 500 m MODIS burned-area product to resolve well. Reliable reference labels are scarce, and labels derived from active-fire detections and spectral rules are noisy, spatially biased and partly circular. We built a weakly supervised workflow that combines Sentinel-2 spectral change, Visible Infrared Imaging Radiometer Suite (VIIRS) active-fire detections and MODIS MCD64A1 as a coarse calibration reference. We compared three pseudo-label recipes, five feature sets and seven classifiers using nested district-held-out cross-validation (three seeds, five folds). The best configuration, a multilayer perceptron trained on the very-strict recipe and the full optical feature set, reached a mean Cohen's kappa of 0.395 (AUC 0.753, F_1 0.747); Random Forest, XGBoost and LightGBM performed about the same. Stricter recipes agreed more with their own held-out pseudo-labels, which says nothing yet about independent map accuracy. For deployment in Google Earth Engine we kept a Random Forest and calibrated its threshold against MODIS, which moved the cutoff from 0.30 to 0.60. At 0.60 the mapped burned area matched MODIS (ratio 1.005), district-level R^2 was 0.636 and the Spearman correlation with district fire counts was 0.779, while pixel-level agreement stayed low (F_1 = 0.205; κ = 0.093). Applied to Haryana without retraining, the model reached κ = 0.641. Punjab kappa fell from 0.662 in 2023 to 0.393 in 2024, and adding Sentinel-1 features did not help. The maps are suited to district-level burden estimates and hotspot screening, not to delineating individual scars. The code and data are publicly available in the GitHub repository.

1 Introduction

Burning crop residue after harvest is common in northwestern India and a long-standing target for satellite monitoring. Punjab's rice–wheat system leaves large amounts of straw, and burning it causes severe short-term air pollution and makes it harder to evaluate mitigation policy (Bhuvaneshwari et al., 2019; Jethva et al., 2019; Lan et al., 2022). Planners need spatially explicit burned-cropland maps for district-level burden estimates, hotspot identification, emissions analysis and the evaluation of residue-management programs. The MODIS MCD64A1 burned-area product gives a consistent regional record, but at 500 m it tends to miss or merge small, fragmented agricultural burns (Giglio et al., 2018; Roteta et al., 2019). Sentinel-2 can resolve these burns at 10–20 m, provided a mapping model can be trained without exhaustive reference labels.

Burned fields are hard to separate from their surroundings because the post-harvest landscape changes quickly. Crop removal, exposed soil, tillage, moisture changes and senescence all produce spectral changes that overlap with those of fire, and burn scars can be patchy and short-lived. Methods developed for wildfire therefore transfer poorly, and cropland-specific context helps (Singh et al., 2009). Manually interpreted burn maps would solve the label problem, but producing them at the scale that supervised learning needs is expensive.

Weak supervision is a practical alternative: labels are built from indirect evidence such as active-fire detections, spectral rules and existing products. The catch is that the label rules shape what the model learns, since the

thresholds, spatial buffers and negative-sampling choices all define the problem. If a burn index used to build the labels is also given to the classifier, the model can simply reproduce the rule, and a good score against held-out weak labels may then reflect the labeling recipe and not burned cropland on the ground. Random train–test splits make this worse, because neighboring pixels are not independent (Karasiak et al., 2022; Roberts et al., 2017).

We study these issues for Punjab and use Haryana only to test geographic transfer. Sentinel-2 supplies the optical features, VIIRS detections supply fire context for building the labels, and MODIS MCD64A1 serves as a coarse external reference for calibrating the decision threshold and checking agreement; it is not pixel-level ground truth. District fire counts give an independent check at aggregate scale. NBR and dNBR, which define the labels, are kept out of the classifier inputs. We ask how much the pseudo-label recipe, the feature set and the classifier each affect district-held-out performance, whether the threshold chosen in cross-validation gives a sensible statewide map or needs separate calibration, and how the selected model behaves in Haryana, in a second year, and with Sentinel-1 features added.

2 Literature Review

Burned-area mapping has moved from coarse global products toward Landsat- and Sentinel-based change detection. MCD64A1 combines spectral change with active-fire data into a consistent 500 m record and is still the main global reference (Giglio et al., 2018). Studies of small fires, however, keep finding that products at this resolution omit or merge burns that 10–30 m imagery can recover. Sentinel-2 algorithms delineate smaller scars using temporal change, region growing or active-fire-seeded candidate areas (Roteta et al., 2019; Sali et al., 2021), and Landsat–Sentinel combinations and cloud-computing workflows extend this to larger areas (Long et al., 2019; Roy et al., 2019). Agricultural fires stand to gain the most, since single fields are often smaller than a MODIS pixel.

Active-fire detections and optical burn scars carry different information, so neither can simply stand in for the other. VIIRS provides frequent thermal detections at a nominal 375 m and is good for locating fire in space and time (Schroeder et al., 2014). Optical imagery has finer spatial detail but needs cloud-free scenes and a spectral change that lasts. Many active-fire-assisted methods therefore use hotspots as seeds or spatial priors and let optical change draw the boundary. In northwestern India, residue burning varies by district, year and management practice, so neither fire counts nor a single coarse product gives complete pixel-level truth (Liu et al., 2020; Singh et al., 2009).

Weak supervision covers learning from heuristic, noisy or incomplete labeling sources when manual labels are not available. Snorkel showed that labeling functions and proxy sources can generate training data at scale, with the usefulness depending on how accurate, independent and complete the sources are (Ratner et al., 2020). A threshold rule on spectral change and distance to active fires is a labeling function, even if it is not called one, and such a rule can already hold much of the predictive signal before any classifier is trained. For this reason we treat label design as an experimental factor and report it, instead of burying it in preprocessing.

Label noise also raises the question of which model and loss to use. Random Forest is popular in remote sensing because it handles nonlinear interactions, correlated predictors and modest sample sizes and is easy to operate (Belgiu and Drăguţ, 2016; Breiman, 2001). XGBoost and LightGBM are efficient boosted-tree alternatives (Chen and Guestrin, 2016; Ke et al., 2017), and multilayer perceptrons test whether learned nonlinear combinations help. Symmetric cross-entropy was proposed to make neural networks less sensitive to noisy labels (Wang et al., 2019), but whether it helps with remotely sensed proxy labels has to be checked case by case.

How we evaluate matters as much as which classifier we use. Spatial dependence between training and test samples inflates accuracy, so geographically separate validation is more realistic for regional mapping (Karasiak et al., 2022; Roberts et al., 2017). Thresholds are a second problem: the cutoff that best matches the weak labels need not give a plausible mapped area or district pattern. External data are needed after model fitting, but each source can only support so much. MODIS agreement can show regional plausibility and calibrate total area, but it cannot confirm 20 m scar boundaries. We are not aware of a study that tests pseudo-label design, circularity-aware features, several classifier families, spatially held-out evaluation, external threshold calibration and transfer together for fine-resolution burned cropland in Punjab. This paper does so.

3 Materials and Methods

3.1 Study Area, Temporal Scope, and Data Roles

The study area is Punjab, India, with neighboring Haryana held back as a transfer region (Figure 1). District boundaries come from FAO GAUL level 2. We restricted label construction, sampling and inference to cropland (ESA WorldCover 2021 v200, class 40; Zanaga et al., 2022), so the task is burned cropland and not fire in general, and every pixel keeps a district identity for the spatial folds and zonal summaries.

The main experiment covers the 2024 post-monsoon burning season. The earlier Sentinel-2 composite spans 20 September–9 October 2024 and the later one 15 October–4 November 2024 (Earth Engine end dates are exclusive). VIIRS detections cover 1 October–29 November, and MODIS comes from the October and November monthly products, so the optical windows are shorter than the reference period. We processed Punjab 2023 the same way for the cross-year comparison. Sentinel-2 surface reflectance is the main optical input (Drusch et al., 2012). Sentinel-1 ground-range-detected imagery is used only in the fusion experiment (Torres et al., 2012). SNPP and NOAA-20 VIIRS detections give the active-fire context, MODIS MCD64A1 is the external calibration and agreement reference, and district fire counts give a rank-correlation check.

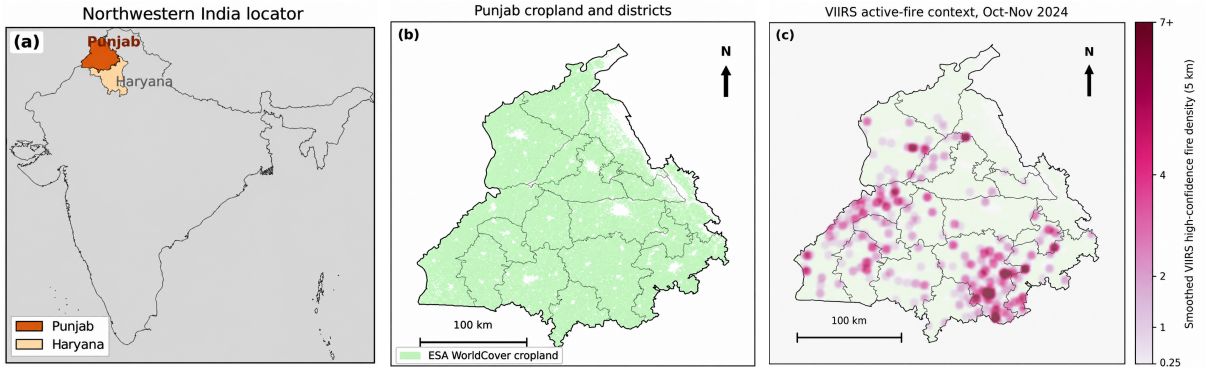


Figure 1: Study area and fire context. (a) Punjab and Haryana in northwestern India. (b) Punjab districts and the WorldCover cropland mask. (c) Density of high-confidence VIIRS detections over Punjab cropland, 1 October–29 November 2024.

3.2 Framework Overview and Image Preprocessing

The workflow has seven steps: seasonal compositing, weak-label generation, quality-control gating, matched sampling, district-held-out evaluation, statewide probability inference with external threshold calibration, and robustness tests (Figure 2). Google Earth Engine handled data assembly, preprocessing, weak-label construction and map export. Python handled classifier benchmarking, paired statistics, calibration analysis and reporting.

We filtered Sentinel-2 scenes by date and area and kept those with `CLOUDY_PIXEL_PERCENTAGE` ≤ 80. We then masked cloud, cirrus, shadow and invalid pixels using a cloud-probability threshold of 35, QA60 and the Scene Classification Layer, and built median composites for the earlier and later windows. We kept ten bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11 and B12) and computed the normalized difference vegetation index (NDVI), normalized difference moisture index (NDMI), normalized difference red-edge index (NDRE) and the earlier-to-later differences. NBR and dNBR were computed for label construction only and are absent from every classifier feature set. The raw SWIR2 change term is $dB12 = B12_{\text{later}} - B12_{\text{earlier}}$. For the fusion test we preprocessed Sentinel-1 GRD scenes (IW mode, descending orbit, VV and VH polarization) into median earlier and later composites and a later-minus-earlier backscatter difference, exported at 20 m. Before training, each recipe had to pass several plausibility checks: at least 95% cropland coverage in each optical composite, at least 1,000 high-confidence VIIRS detections, a nonzero MODIS burned reference, and at least 10,000 statewide positive weak labels. Weak positives also needed an overlap score of at least 0.25 with MODIS at 500 m support, and we screened district label availability and sampled feature correlations.

3.3 Weak-Label Recipes and Feature Design

Weak labels combine VIIRS fire proximity with burn-consistent optical change. We merged high-confidence SNPP and NOAA-20 detections over the VIIRS window, converted them into a fire-context layer, and computed the distance to the nearest detection. Figure 3 shows the construction.

Every cropland pixel became a weak positive, a weak negative or unlabeled. A positive had to lie within a recipe-specific distance of a fire and pass dNBR and SWIR-change gates; the most selective recipe also required an NDVI decrease. A negative was either far enough from any detected fire or near a fire but spectrally quiet. Pixels that met neither rule were left out of training.

The three recipes form a gradient from quantity to selectivity (Table 1). The moderate recipe has the widest fire buffer and the lowest dNBR threshold. The strict recipe tightens both the spatial and the spectral conditions. The very-strict recipe cuts the near-fire radius to 1.5 km, raises the dNBR and $dB12$ requirements, pushes the far-from-fire distance out, lowers the hard-negative limits and adds the NDVI-decrease gate. All three use the same source layers and seasonal windows; only thresholds, buffers and gates differ.

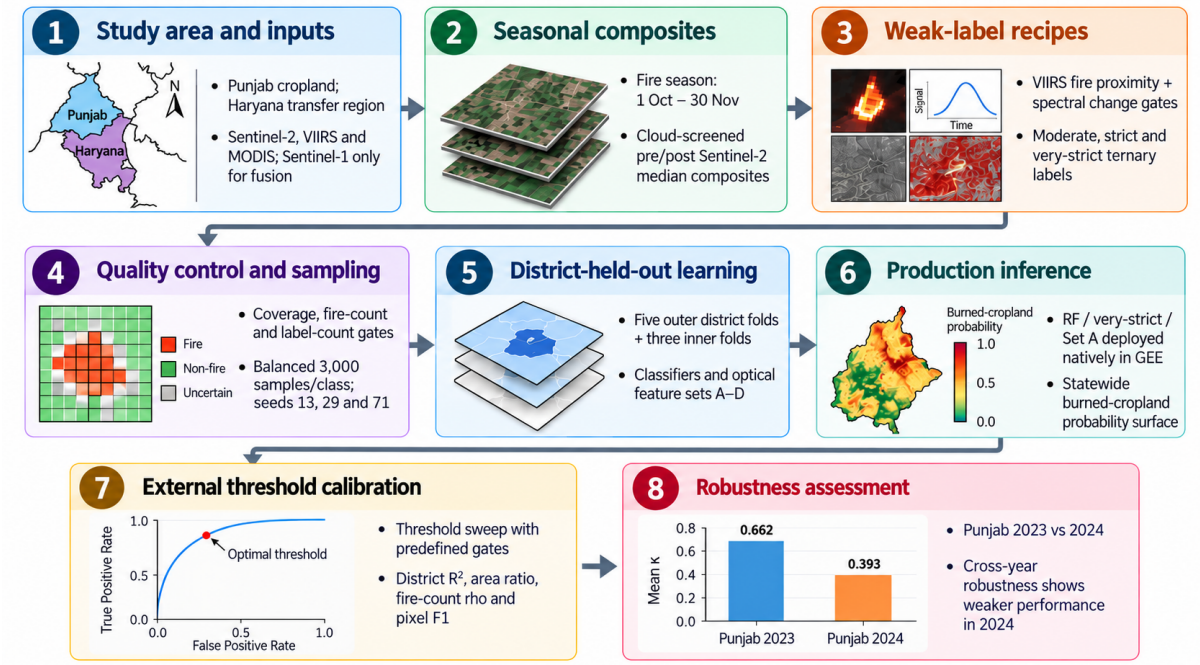


Figure 2: Overview of the workflow. Sentinel-2 preprocessing, VIIRS-anchored pseudo-labels, district-held-out evaluation, statewide RF inference, threshold calibration against external data and robustness tests are separate stages. In the schematic, pre/post means the earlier/later optical composite, and October–November is the full reference season. NBR and dNBR build the labels but are never classifier inputs.

We defined five feature sets by purpose (Table 2). Set A is the full 39-variable optical set with earlier, later and delta values. Set B keeps only the 13 delta variables. Set C combines earlier state with deltas. Set D drops the SWIR-related predictors and serves as the main circularity ablation. Set E adds six Sentinel-1 variables to Set A. Because Set A was exported as a superset, we rebuilt Sets B–D by selecting columns from the same samples.

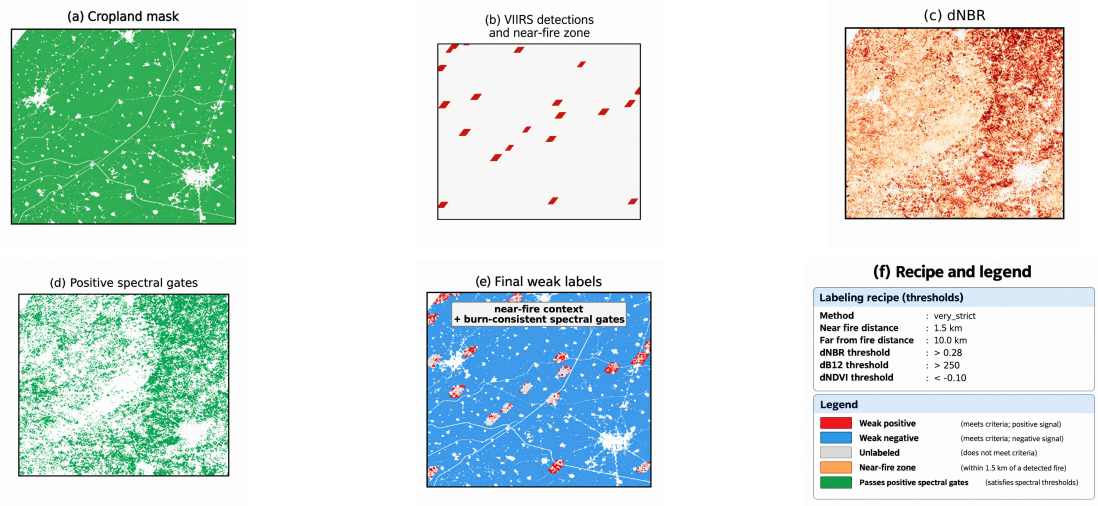


Figure 3: Weak-label construction. Cropland and VIIRS fire context are combined with spectral gates to give weak positives, weak negatives and unlabeled pixels. The parameter block shows the very-strict recipe.

Recipe	d NBR	Near km	*dB* 12	HN d	HN b	Far km	NDVI
Moderate	0.18	3.0	> 200	0.06	120	8	None
Strict	0.22	2.0	> 200	0.05	100	8	None
Very-strict	0.28	1.5	> 250	0.04	80	10	> 0.10

Table 1: Weak-label recipe parameters. Near- and far-fire distances are in km. HN = hard negative; *d* and *b* are the dNBR and *dB*12 limits. The recipes use the same inputs and seasonal windows and differ only in thresholds and gates.

Set	Size	Composition	Purpose
A	39	Earlier, later, and delta values for 10 Sentinel-2 bands plus three indices	Full optical set used in production
B	13	Delta-only optical variables	Tests whether change alone is sufficient
C	26	Earlier optical state plus delta variables	Earlier state plus change, fewer redundant terms
D	33	Full optical set without SWIR-related predictors	Circularity ablation
E	45	Set A plus Sentinel-1 VV/VH earlier, later, and delta	Simple optical–radar fusion test

Table 2: Feature-set definitions. NBR and dNBR are excluded from every feature set because they are used to build the labels.

3.4 Sampling, Classifiers, Baselines, and District-Held-Out Evaluation

For each recipe and seed we drew a stratified sample of 3,000 weak positives and 3,000 weak negatives, using three fixed seeds (13, 29 and 71). We assigned whole districts, not pixels, to five outer folds, so that all samples from a district are held out together. This gives 15 seed–fold evaluations per configuration, limits local spatial leakage and measures transfer to unseen districts. Within each outer training set, a three-fold district-based inner validation chose a probability threshold from 0.1 to 0.9 in steps of 0.1 by maximizing Cohen's kappa. Held-out districts played no part in tuning the threshold.

We compared Logistic Regression, Random Forest (RF), Gradient Boosted Trees, XGBoost, LightGBM, a multilayer perceptron (MLP) and an MLP trained with symmetric cross-entropy (MLP-SCE). RF used 300 trees, square-root feature subsampling and a minimum leaf size of 1. Gradient Boosted Trees used 300 estimators, a learning rate of 0.1 and at most 64 leaf nodes. The MLP had hidden layers of 128 and 64 units with batch normalization, ReLU activation, dropout of 0.3, Adam optimization (learning rate 0.001), batch size 256 and 200 epochs. MLP-SCE used the same architecture with symmetric cross-entropy weights $\alpha = 1$ and $\beta = 1$. All classifiers ran on the same matched samples and district folds. Our two non-learned baselines were the recipe-rule threshold and a fixed dNBR threshold of 0.12. We did not use MODIS as a classifier baseline because we did not complete a valid district-held-out MODIS benchmark.

Internal evaluation scores predictions against held-out weak pseudo-labels, not human-labeled pixels. Each recipe defines its own evaluation labels and sampled population, so differences between recipes do not isolate label quality against a common reference. Cohen's kappa was the main ranking metric, alongside F_1, balanced accuracy, area under the receiver operating characteristic curve (AUC), precision, recall, specificity, the selected threshold and the predicted positive rate. For the key paired comparisons we used Wilcoxon signed-rank tests across matched seed–fold splits with Bonferroni correction, and Cliff's delta.

3.5 Production Model and External Deployment-Threshold Calibration

We kept the best trained model and the production model separate. The best learned configuration was MLP/very-strict/Set A. We deployed RF/very-strict/Set A instead because the mapping workflow builds the model directly in Google Earth Engine, whose exporter supports native RF and gradient-boosted trees but not a PyTorch MLP. The MLP's advantage over RF on Set A was only 0.00253 kappa and not significant after correction (corrected $p = 1.0$; Cliff's $\delta = -0.013$), so using RF cost us little.

Set B did slightly better than Set A for RF in cross-validation and is more compact. We still used Set A in production because it had already been fixed, exported and taken through the whole statewide calibration workflow. We therefore report Set A as the deployed configuration, not as the best RF feature set.

We retrained RF/very-strict/Set A on all labeled Punjab samples and evaluated the statewide probability surface at 17 candidate thresholds from 0.10 to 0.90 in steps of 0.05. Three deployment gates applied: district-level R^2 against MODIS of at least 0.50, a mapped-to-MODIS burned-area ratio of at most 2.0, and a district fire-count Spearman correlation of at least 0.60. We ranked thresholds by (1) the number of gates passed; (2) passing the area-ratio gate; (3) area ratio closest to 1 on a log scale; (4) higher district R^2; (5) higher Spearman correlation; (6) higher pixel-level MODIS F_1; and (7) lower threshold. For the pixel-level checks we aggregated the 20 m model output to the 500 m MODIS grid before computing overlap, to avoid matching pixels across different supports. Thresholds 0.55, 0.60, 0.65 and 0.70 passed all three gates, and we selected 0.60 because its area ratio (1.005) was closest to parity. Because the MODIS and fire-count values drove this choice, they are calibration statistics, and the pixel metrics are agreement diagnostics against the calibration reference, not independent validation.

3.6 Robustness Experiments

For geographic transfer we fitted three RF/very-strict/Set A models on Punjab samples, one per seed, and applied them directly to Haryana, with no Haryana training, fine-tuning or threshold calibration. Each model used its own Punjab inner-cross-validation threshold (0.30, 0.40 and 0.30). We report the mean sample-level zero-shot kappa over the three models.

For temporal stability we compared Punjab 2023 with Punjab 2024 under the same seasonal design and evaluation; only these two years are reported. For sensor augmentation we compared optical Set A with Set E, which adds earlier, later and delta VV/VH features by simple concatenation. The result applies to this fusion design only, not to radar for burned-area mapping in general.

4 Results

4.1 Recipe Selectivity and Held-Out Pseudo-Label Agreement

Stricter recipes agreed more with their own held-out pseudo-labels. Averaged over classifiers and feature sets, the moderate recipe (81,471 positive labels) gave kappa 0.216 and AUC 0.658. The strict recipe (73,746 positives) gave 0.272 and 0.685. The very-strict recipe had the fewest positives (59,063) but the highest scores, kappa 0.375 and AUC 0.738 (Figure 4). Each recipe is scored on its own labels, so these gains show agreement with the recipe, not better burned-area accuracy.

The best configuration was MLP/very-strict/Set A, with mean kappa 0.3951, AUC 0.753, F_1 0.747 and balanced accuracy 0.703 over 15 district-held-out splits (Table 3). Several others were essentially tied: RF/very-strict/Set B (kappa 0.395), XGBoost/very-strict/Set A (0.394), RF/very-strict/Set A (0.3926) and the top LightGBM variants. A paired comparison of MLP/A against RF/A gave $\Delta\kappa = 0.00253$, corrected $p = 1.0$ and Cliff's $\delta = -0.013$, so the MLP has no real advantage.

The non-learned baselines show how much signal the labeling step already carries. A fixed dNBR threshold stayed weak, at kappa of about 0.126–0.127 for all recipes. The recipe rule, which is close to how the pseudo-labels were built, improved with selectivity and reached kappa 0.426 for the very-strict recipe. It is not a learned model, and its score reflects agreement with the recipe's own labels. What the learned models add is a probability surface that transfers to new areas and a way to combine several change features.

NBR and dNBR were not model inputs, yet removing the SWIR predictors in Set D lowered performance. We read this as evidence that burn-sensitive optical information beyond those two indices contributes to the discrimination. Set B (deltas only) did well with RF, so temporal change carries much of the signal, although we kept the full Set A in production for continuity. These ablations support a multivariate optical-change reading but say little about the importance of any single variable.

Rank	Recipe	Model	Set	Mean κ	SD κ	AUC	F_1	Bal. acc.
1	Very-strict	MLP	A	0.395	0.049	0.753	0.747	0.703
2	Very-strict	RF	B	0.395	0.053	0.747	0.744	0.703
3	Very-strict	XGBoost	A	0.394	0.053	0.740	0.747	0.703
4	Very-strict	RF	A	0.393	0.056	0.744	0.746	0.701
5	Very-strict	LightGBM	B	0.393	0.052	0.740	0.748	0.701

Table 3: Leading learned configurations. Values are means across 15 district-held-out evaluations (three seeds × five folds).

4.2 External Threshold Calibration and Scale-Dependent Agreement

The threshold chosen in cross-validation did not work for the whole state. For RF/very-strict/Set A the median cross-validated threshold was 0.30, and applying it to the full probability map gave a burned area far above MODIS. Calibration moved the threshold to 0.60 (Figure 5). Telling weak-label samples apart and calibrating mapped extent are therefore two separate problems.

Thresholds from 0.55 to 0.70 met all three deployment gates. The ranking rule picked 0.60 because its area ratio of 1.005 was closest to parity. At 0.60, district-level R^2 against MODIS was 0.636 and the Spearman correlation with district fire counts was 0.779 (Table 4). These values show that the map matches MODIS at district scale and ranks districts consistently with fire counts. Since we used them to choose the threshold, they are calibration statistics and not a separate test.

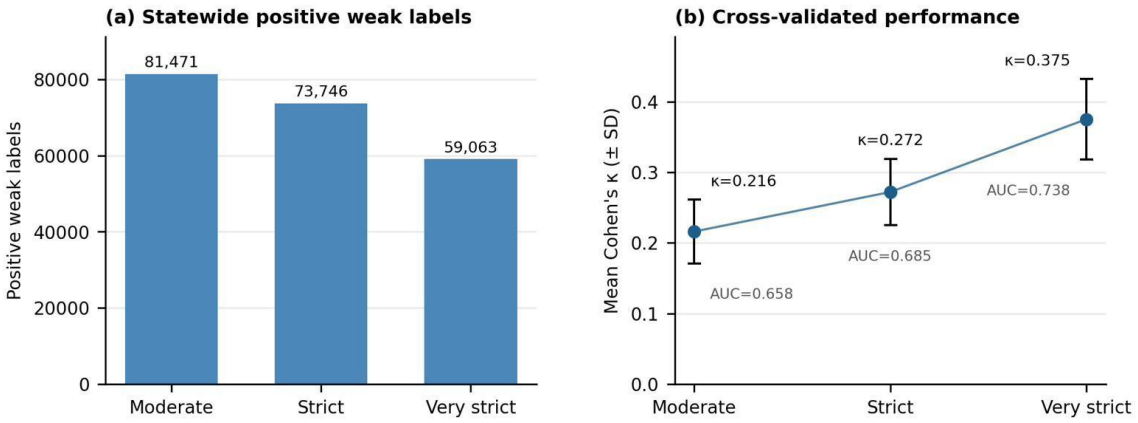


Figure 4: Number of positive weak labels and mean district-held-out agreement for each pseudo-label recipe. The very-strict recipe has the fewest positives and the highest agreement with its own pseudo-labels. Each recipe is scored against its own labels, so the bars are not directly comparable as accuracy.

At the pixel level, agreement with MODIS stayed low: F_1 was 0.205 and kappa 0.093, with precision and recall both near 0.20 and much higher specificity. A 20 m map compared against a 500 m reference will differ because of the mismatch in support, and both products have their own errors. The result does not support claims about exact scar boundaries.

District residuals had a spatial pattern. Firozpur and Moga agreed closely with MODIS, while Sangrur, Muktsar, Bathinda and Mansa were underestimated, and several districts with little MODIS burning were overestimated. The map thus reproduces statewide extent and much of the district ranking but keeps district-specific bias.

Item	Value
Production classifier	Random Forest (300 trees)
Pseudo-label recipe	Very-strict
Feature set	A (39 optical variables)
Median internal CV threshold	0.30
Externally calibrated threshold	0.60
District R^2 vs. MODIS	0.636
Mapped/MODIS burned-area ratio	1.005
Spearman ρ vs. district fire counts	0.779
Pixel F_1 vs. MODIS	0.205
Pixel κ vs. MODIS	0.093
Deployment gates passed	3/3

Table 4: Production configuration and calibration statistics at the selected threshold.

4.3 Geographic Transfer, Cross-Year Stability, and Sentinel-1 Fusion

Zero-shot transfer to Haryana worked well. The three Punjab-trained RF models, applied without Haryana training or recalibration and using the Punjab-only thresholds of 0.30, 0.40 and 0.30, reached a mean sample-level kappa of 0.641. The decision surface thus carried over to a neighboring agricultural region under the same season and sensors, though this does not show wider geographic generalization.

Temporal behavior was much weaker. Kappa was 0.662 for Punjab 2023 and 0.393 for 2024, with a two-year standard deviation of 0.135. We cannot call the framework temporally invariant. Differences in fire intensity, crop timing, atmospheric contamination, composite quality or pseudo-label fidelity could all play a part, and our experiments do not separate them.

Concatenating Sentinel-1 features did not improve on the optical baseline: RF/very-strict/Set A reached a mean kappa of 0.393 and Set E 0.378, a difference of −0.015 (Table 5). This holds only for the earlier/later/delta VV/VH concatenation we tested.

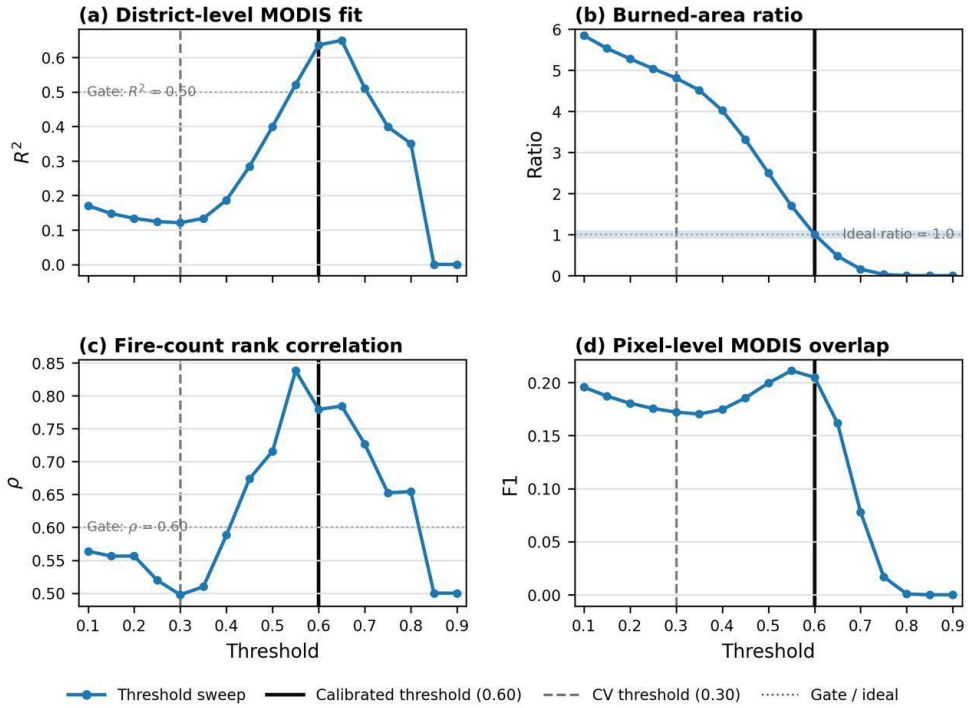


Figure 5: Threshold calibration for RF/very-strict/Set A. The internal threshold (0.30) and the calibrated one (0.60) are compared on district MODIS fit, area ratio, fire-count rank correlation and pixel MODIS F_1. We chose 0.60 with the predefined ranking rule, not by maximizing any single metric.

Experiment	Configuration/protocol	Mean κ	Interpretation
Haryana zero-shot	Punjab-trained RF/very-strict/A; thresholds 0.30, 0.40, 0.30; no Haryana fitting or calibration	0.641	Transfers to the neighboring state
Punjab 2023	Same seasonal framework	0.662	Higher in 2023
Punjab 2024	Primary experiment	0.393	Lower in 2024
S1 + S2 fusion	Set E minus optical Set A	$-0.015\,\Delta\kappa$	No gain from concatenation

Table 5: Robustness and transfer results. Cross-year results cover Punjab 2023 and 2024 only.

5 Discussion

5.1 Interpreting Recipe-Specific Pseudo-Label Agreement

The rise in agreement from the moderate to the very-strict recipe is the main pattern in our results, but it needs careful reading. Tighter fire proximity, stronger spectral gates and harder negatives change both the training labels and the evaluation set, so a higher score may simply mean an easier, more separable problem. We did not measure label purity, and the recipes were not scored against a common independent reference, so the scores cannot tell us whether the labels or the maps became more accurate.

The near-tie between MLP, RF, XGBoost and LightGBM under the very-strict recipe matters in practice. A flexible classifier cannot recover information that the weak labels lack or contradict. The recipe-rule score points the same way: much of the discrimination was already in the proxy rules. What learning added was generalization beyond hard rule boundaries, a combination of several change features and a continuous statewide score. The ablations fit post-harvest burning. The drop without SWIR predictors and the strength of the delta-only features are consistent with abrupt changes in vegetation condition, moisture and shortwave-infrared response. These changes are not specific to burning, since harvest, tillage, moisture change and atmospheric effects can look similar.

5.2 Why External Calibration Was Necessary and What Scale It Supports

The gap between the cross-validated threshold (0.30) and the deployed one (0.60) shows that weak-label performance does not carry over to mapping directly. Cross-validation picked the cutoff that best separated weak positives from

weak negatives in held-out districts, but those classes came from selective rules and do not reflect the prevalence or uncertainty across all cropland pixels. Statewide inference needed its own threshold choice, based on mapped area and district patterns.

Calibration made aggregate extent and district ranking more plausible, but MODIS is not ground truth. MCD64A1 has coarser support and its own omission and commission errors, especially for small agricultural burns. District R^2, area ratio and fire-count correlation therefore say more about regional burden than pixel overlap does. The calibrated map suits district comparison, hotspot screening and prioritization. It cannot yet give exact field boundaries, individual scars or burn timing.

5.3 Transferability, Circularity, and Limitations

The Haryana result suggests that the optical-change representation is not tied to the Punjab district partition used for development. Haryana is adjacent, though, with similar crop calendars and landscapes, so this is a narrow test. The year-to-year result is more worrying. A model that transfers to a neighboring state in the same season can still change a lot between years, because atmospheric conditions, residue management, crop phenology and label coverage all vary.

We reduced circularity but did not remove it. Leaving out NBR and dNBR stopped the classifier from reusing the main label threshold directly, and the no-SWIR ablation tested reliance on related bands. The remaining optical predictors still respond to the same burn process as the label variables. That is reasonable for prediction, but it means that weak-label metrics cannot be read as independent pixel accuracy.

The optical windows do not cover every fire in the reference period. Early burns may already appear in the earlier composite, and scars made after the later composite ends on 4 November are not seen, even though VIIRS runs to 29 November and MODIS includes the full October and November products. This mismatch limits how the agreement with the seasonal references can be interpreted.

Other limitations are the lack of an independent human-labeled pixel reference, the heuristic weak labels, the coarse calibration data, the spatially structured district residuals, the weak 2023–2024 stability, and a negative Sentinel-1 result for only one simple fusion design. Future work should add independent field or visual reference samples, year-adaptive labels and thresholds, uncertainty maps, and better temporal or multimodal fusion. For now, the fair claim is calibrated regional monitoring under label scarcity, not proven scar delineation.

6 Conclusion

We developed a weakly supervised workflow for mapping post-harvest burned cropland that treats label construction, district-held-out learning and map-level threshold calibration as separate steps. The most selective recipe agreed best with its own pseudo-labels, and MLP/very-strict/Set A had the highest kappa (0.395), with RF, XGBoost and LightGBM close behind. We deployed RF/very-strict/Set A because Google Earth Engine supports it natively and it loses little to the MLP. Calibrating its threshold to 0.60 gave a district R^2 of 0.636, an area ratio of 1.005 and a fire-count Spearman correlation of 0.779, while pixel-level agreement with MODIS stayed low. The maps are therefore useful for district burden and hotspot ranking, not for field boundaries or year-to-year comparison. Transfer to Haryana worked well, but Punjab results differed sharply between 2023 and 2024, and simple Sentinel-1 concatenation added nothing. Independent reference samples are needed before exact scar delineation can be claimed.

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